THE INFOMEDIARY — PASS B DRAFT

Part II · Part III · Part IV (Pass B addendum to Pass A FINAL v1.2)

Document ID: ERES-INFOMEDIARY-BOOK-2026-001-v1.3-PASS-B-DRAFT **Author:** Joseph Allen Sprute (ERES Maestro) **Affiliation:** ERES Institute for New Age Cybernetics, Bella Vista, Arkansas, USA **ORCID:** 0000-0001-9946-3221 **Document Class:** NAC Foundational Instrument Specification (Pass B draft, addendum) **Date:** May 9, 2026 **License:** CARE Commons Attribution License v2.1 (CCAL v2.1)

Composes on:

- ERES-INFOMEDIARY-BOOK-2026-001-v1.2-PASS-A-FINAL — Front Matter, Part Zero, Part I
- ERES-INFOMEDIARY-LOCK-MEMO-2026-001-v1.2 — canonical lock authority

Pass B contents (this addendum):

- **Part II** — The Three Volumes in Depth (Chapters 4, 5, 6)
- **Part III** — The Matrix Walk (twelve cells × three entry surfaces, plus synthesis)
- **Part IV** — DATABANK Architecture under DOFA Governance

Integration plan: Pass B remains a DRAFT addendum until the v2.0 release sweep merges it with Pass A FINAL v1.2 (and any Pass C / Pass D content) into a single canonical Volume Zero. No part of this addendum modifies Pass A material; Pass B extends only.

Lock inheritance: All three lock boxes from Memo v1.2 (A: Underwriting Direction; B: Five-Layer Non-Extensibility; C: Pre-Qualification Inversion) and the SEPLTA "A" = Administrative correction govern this addendum without restatement.

Part II — The Three Volumes in Depth

Part II takes each Trilogy volume and unfolds its canonical formulation operationally. The three chapters are not symmetric in length: Security-Clearance carries the most because it is doubled (operational + governance frames); Data-Integrity carries the second-most because each of its five layers requires its own treatment; One-Good is the lightest because its operational frame (RT lookup for Adaptive-Fixed Q/A) is the most directly mappable to existing infrastructure.

The chapters follow a common pattern:

1. The volume's canonical formulation, restated from Memo v1.2 §1

2. Operational unfolding of the formulation
3. The four cells (one per Pillar) with operational specifications
4. Failure modes
5. Relationship to EAAP-POL

Chapter 4 — One-Good

§4.1 Canonical Formulation

> **One-Good** = need RT lookup for Adaptive-Fixed QuestionAnswer == HowWay MyWay $IT === UBIMIA Need-to-know for Common Core Sustenance

Three equivalence tiers. The operational frame (`=`) is real-time lookup for Adaptive-Fixed Question/Answer. The structural equivalence (`==`) is the HowWay/MyWay/$IT triadic pattern. The identity-level definition (`===`) is UBIMIA Need-to-know for Common Core Sustenance.

§4.2 The RT Lookup Architecture

Real-time lookup means: the Civigen submits a query and receives Sim OUTPUT within a latency envelope that supports operational decision-making. RT-tier responsiveness is not "fast"; it is **fast enough that the answer arrives before the situation it concerns has changed.** The latency envelope is therefore situational: a query about whether to take an immediate medical action requires sub-second response; a query about whether to plant a particular crop next season requires only sub-minute response. RT is calibrated to the situation, not absolute.

The RT requirement is canonical to One-Good because Common Core Sustenance — the volume's identity-level commitment — is operationally meaningful only when sustenance support arrives in time. A Common Core Sustenance system that produces correct answers an hour after the situation has resolved is a research instrument, not an Instrument-of-Faith. One-Good's RT commitment is what distinguishes it from non-RT lookup systems.

The Adaptive-Fixed compound is structural: every One-Good query has both an *adaptive* component (responsive to the Civigen's situation, location, capacity, Pillar selection) and a *fixed* component (anchored to canonical UBIMIA-Manual material on what Common Core Sustenance includes, what BERA dimensions apply, and what EarnedPath actions are constitutionally available). Neither component alone is sufficient. Pure adaptive response would untether from canonical commitments; pure fixed response would fail to serve the actual Civigen. The compound *is* the volume's operational signature.

§4.3 The HowWay / MyWay / $IT Triadic Pattern *(structural equivalence)*

The structural equivalence — `== HowWay MyWay \$IT` — names the three poles around which One-Good's Q/A architecture organizes:

- **HowWay** — the process/method tier. *How* is the question being answered? What is the canonical method? HowWay is the fixed pole — the way the architecture answers questions of this kind, anchored to the UBIMIA-Manual specification. HowWay is reusable across Civigens; it is the common method.
- **MyWay** — the personal tier. *My* path through the answer. MyWay is the adaptive pole — the answer particularized to the Civigen's situation, capacity, and selection profile. MyWay is non-reusable across Civigens; it is what makes the answer *the Civigen's answer*.
- **$IT** — the concrete tier. The *it*, the embodied output, the actionable Sim OUTPUT in the Civigen's hand. $IT is what HowWay × MyWay produces — the specific deliverable, the specific EarnedPath action set, the specific BERA reading, the specific covenantal language. $IT is the round-trip's external closure.

The triad is canonical because Common Core Sustenance answers must be all three — methodologically grounded (HowWay), personally appropriate (MyWay), and concretely actionable ($IT). A One-Good answer missing any pole is incomplete.

§4.4 UBIMIA Need-to-Know for Common Core Sustenance *(identity-level)*

The identity-level definition specifies *what* One-Good is. It is the Trilogy volume that operationalizes the architecture's commitment to **common-core sustenance** — the basic-needs floor below which no Civigen falls — through **need-to-know** access semantics.

"Need-to-know" here is not the security-tier need-to-know that Security-Clearance operationalizes. It is the **sustenance-tier** need-to-know: the principle that every Civigen needs to know what Common Core Sustenance is available to them, how to access it, what it depends on, and how to maintain it. This need-to-know is *expansive* (every Civigen has it) rather than *restrictive* (only some Civigens have access) — opposite to Security-Clearance's restrictive need-to-know. The doubled use of "need-to-know" across the two volumes is intentional: One-Good's expansive form and Security-Clearance's restrictive form are complementary commitments, not redundant.

Common Core Sustenance is specified in the UBIMIA Manual v1.1 across the four Constitutional Pillars and is canonically not re-specified here; this Volume cites the Manual as binding.

§4.5 The Four One-Good Cells

The four cells produced by One-Good × {HELP, USE, ENERGY, LAW}:

One-Good × HELP — Sustenance Care. RT lookup of available sustenance resources within reach (food, medical, shelter, basic care). The cell where a Civigen's most immediate sustenance needs are addressed. HowWay grounds the answer in UBIMIA Manual sustenance specification; MyWay particularizes to the Civigen's location and capacity; $IT delivers the concrete sustenance access path.

One-Good × USE — Adaptive Q&A. RT operationalization of the HowWay/MyWay/$IT triadic Q/A pattern itself. The cell where ordinary day-to-day Civigen questions are answered. The cell most likely to be selected by single-cell Civigens new to the architecture.

One-Good × ENERGY — Resource Sufficiency. RT accounting of energy and resource availability against the Civigen's commitment surface. The cell where energetic budgeting, resource-flow tracking, and sufficiency-versus-scarcity diagnosis happen. Particularly important at THOW-tier where energetic margins are narrow.

One-Good × LAW — Covenant of Provision. RT verification of legal commitments to provide. The cell where a Civigen confirms their constitutional rights to common-core sustenance, where institutions confirm their obligations, and where the covenant between Civigen and architecture is made operationally visible. The cell that grounds the architecture's claim to legitimacy in the legal register.

§4.6 Failure Modes

One-Good fails in four canonical ways:

1. **RT latency exceeds the situation's threshold.** The system returns a correct answer too late to be operationally useful. Diagnostic: situation-specific latency comparison against a calibrated envelope.
2. **Adaptive component over-fits to context, losing canonical anchoring.** MyWay drifts away from HowWay; the answer is personalized but no longer methodologically grounded. Diagnostic: HowWay-anchoring check on Sim OUTPUT.
3. **Fixed component locks against legitimate adaptation.** HowWay refuses to particularize to MyWay; the answer is methodologically pure but operationally misaligned. Diagnostic: MyWay-fit check on Sim OUTPUT.
4. **Common Core boundary disputed.** What counts as "core" is contested between Civigen and architecture. Diagnostic: UBIMIA Manual citation alignment.

Failure modes 2 and 3 are mirror images and are diagnosed together. Failure mode 4 is the most consequential; resolution requires DOFA adjudication, not within-round-trip correction.

§4.7 Relationship to EAAP-POL

EAAP-POL constraints on One-Good are minimal because the volume's commitments are expansive: every Civigen at any tier may select One-Good cells. The only EAAP-POL constraint is that selection of One-Good × LAW (Covenant of Provision) requires the Civigen to be a recognized Civigen under DOFA registration — not a Class A/B distinction, but a registration check. Sustenance-tier need-to-know is universal among registered Civigens.

Chapter 5 — Security-Clearance

§5.1 Canonical Formulation (Doubled)

Security-Clearance carries two equally canonical formulations from Memo v1.2 §1.2.

Operational frame:

> **Security-Clearance** = need-to-know == Defense-in-Depth (DID) === FAVORS Def-Rel (Hand vs Head) for S3C SLA Underwriting (Infomediary underwritten / Infomediator underwrites)

Governance frame:

> **Security-Clearance** = CBGMODD == CyberRAVE Def-Rel === FAVORS Commit to SROC (or similar EarnedPath)

The doubling is canonical, not optional. A User-GROUP selecting Security-Clearance cells commits to both frames. Selection of either alone is incoherent because the operational frame describes *how* the volume functions while the governance frame describes *why*; one without the other is mechanism without authorization or authorization without mechanism.

§5.2 Operational Frame Unfolded

Need-to-know (restrictive form). Unlike One-Good's expansive need-to-know, Security-Clearance's need-to-know restricts access to those whose role, station, or attestation justifies access. The principle is straightforward: Civigens see what they need to see to do what they have committed to do, and no more. Access without need is a vulnerability; access with need is a tool.

Defense-in-Depth (DID). DID is the multi-layer security tiering that ensures no single failure compromises the round-trip. The architecture treats security not as a perimeter to be defended but as a series of nested confirmations, each of which can fail without bringing down the whole. DID is canonical because the Infomediary's underwriting role (Lock Box A) makes single-point-of-failure architectures unacceptable — an underwritten asset cannot afford catastrophic failure modes.

FAVORS Def-Rel (Hand vs Head). FAVORS is the architecture's resource-allocation layer; Def-Rel is its Defense-Relational dimension; Hand-vs-Head is the canonical split:

- **Hand-tier** authorization is physical, manual, embodied — a signature, a physical key, a face-to-face confirmation. Hand-tier is appropriate for situations where the cost of a wrong answer is bounded and where embodiment is valuable.
- **Head-tier** authorization is cryptographic, computational, abstract — a key signature, a zero-knowledge proof, a multi-party computation. Head-tier is appropriate for situations where cost is unbounded or where embodiment is impractical (cross-jurisdictional, archival, federation-scale).

Most operational situations require *both* — Hand-tier for the immediate transaction and Head-tier for the audit trail and federation propagation. The Hand-vs-Head split is therefore not a choice but an allocation; Def-Rel determines the allocation per situation.

S3C SLA Underwriting. S3C (Solid-State Smart-City) is the deployment scale at which Security-Clearance's underwriting commitments become legible at city-tier. SLA (Service Level Agreement) is the contractual instrument through which underwriting is specified. The Infomediary, as underwritten asset, carries the SLA's risk; the Infomediator, as underwriting agent, carries the SLA's authority. Risk flows down (Infomediator → Infomediary, per Lock Box A); authority flows up (Infomediary → Infomediator → DOFA → GAIA).

§5.3 Governance Frame Unfolded

CBGMODD seven-seat council. Citizen, Business, Government, Military, Ombudsman, Dignitary, Diplomat. Seven seats, each carrying a distinct governance perspective and a distinct default selection profile. The council is DOFA's operational expression; Security-Clearance's governance frame routes through CBGMODD.

The seven seats are not symmetric. Citizen and Business represent the two largest population classes; Government and Military represent the two largest authority structures; Ombudsman represents the audit/redress function; Dignitary represents the spiritual/cultural register; Diplomat represents the cross-federation interface. A query that requires Security-Clearance × LAW (underwriting authority) must touch all seven seats in some configuration; lighter Security-Clearance queries may touch fewer seats but never zero.

CyberRAVE Def-Rel. CyberRAVE — *Cybernetic Ratings Abolishing Veiled Exchanges* — is the rating substrate that makes Security-Clearance's Def-Rel allocations honest. Without CyberRAVE, Hand-vs-Head allocation could drift toward whatever is convenient; with CyberRAVE, the allocation is rated, attested, and audit-trailed. CyberRAVE Def-Rel is what prevents Security-Clearance from collapsing into security-theater.

FAVORS Commit to SROC (or similar EarnedPath). SROC — Smart-Resonant Offset Contracts — is the canonical EarnedPath instrument through which Security-Clearance's governance commitment is operationally expressed. SROC instruments are contracts that:

- Encode security commitments at the cell level (per Pillar)
- Resonate with BERA RATE composites (per the Memo v1.2 §1.4 isomorphism)
- Offset risk through underwriting (per Lock Box A)

"Or similar EarnedPath" preserves architectural openness — Security-Clearance does not foreclose alternatives to SROC — but Pass B specifies the formal designation process for any alternative.

§5.4 Formal Designation Process for Alternative EarnedPath Instruments

An EarnedPath instrument other than SROC is admissible to Security-Clearance's governance frame only after passing all five gates:

1. **Need-criteria gate.** The proposed instrument must address a need that SROC cannot or does not address (a coverage gap, a jurisdiction-specific constraint, a scale boundary).
2. **DOFA ratification gate.** The seven-seat CBGMODD council must ratify by supermajority (5/7 minimum) the inclusion of the alternative instrument.
3. **EAAP-POL alignment gate.** The instrument's commitments must align with EAAP-POL constraints — no instrument may grant access the policy framework forbids.
4. **Pilot deployment gate.** A bounded pilot deployment (typically THOW or HFVN scale) must demonstrate operational viability before full designation.
5. **VERTECA federation gate.** The instrument must be expressible in the License Example Mirror format such that other federation instances can adopt or reject it cleanly.

Five gates, sequential, no skipping. The process is deliberately heavy because the cost of admitting a defective alternative EarnedPath instrument is borne by every Civigen whose Security-Clearance selection profile activates against the defective instrument.

§5.5 The Wielder/Instrument Financial Doublet (Lock Box A)

Lock Box A from Memo v1.2 §1.2 is restated here at full operational depth:

> The Infomediary is always the underwritten asset.
> The Infomediator is always the underwriting agent.
> Financial risk flows downward (Infomediator → Infomediary).
> Authority flows upward (Infomediary → Infomediator → DOFA → GAIA).

The doublet is what makes Security-Clearance's SLA underwriting tractable. An Infomediary without an Infomediator's underwriting carries unbounded risk — a defective round-trip has no upstream party to absorb the consequences. An Infomediator without an Infomediary to underwrite has no instrument to wield. The doublet pairs them as a single financial unit.

Operationally, this means every Security-Clearance round-trip has an explicit underwriting record: which Infomediator underwrites this Infomediary at this round-trip, for what cells, against what BERA RATE thresholds, under what SLA. The record is attested under EAAP and revocable under EAAP-CRL. If the Infomediator's underwriting is revoked, the Infomediary is offline for Security-Clearance purposes until a new Infomediator underwrites it. There is no "default" or "self-underwriting" mode; Security-Clearance is always doubly-named.

§5.6 The Four Security-Clearance Cells

Security-Clearance × HELP — DID for Vulnerable. Defense-in-Depth applied to vulnerable populations: children, elderly, disabled, refugees, those with reduced capacity for self-defense.

The cell where the architecture's protective commitments are most strongly enforced. Hand-vs-Head allocation skews to Hand-tier (embodied protection) because vulnerable populations often cannot operate Head-tier authorization independently.

Security-Clearance × USE — Hand/Head Purchase Tier. Authorization tiering for transactional use. Hand-tier for routine transactions (under threshold); Head-tier for cryptographic-scale (above threshold or cross-jurisdictional). The cell most directly engaged in commercial activity.

Security-Clearance × ENERGY — Grid Access Tier. Authorization tiering for energy-grid access. THOW-tier (single-household micro-grid clearance), HFVN-tier (village-network grid clearance), S3C-tier (city-grid clearance), FDRV-tier (closed-grid clearance). The cell that scales across the deployment-scale spectrum (per Pass A FINAL §0.6.1).

Security-Clearance × LAW — Underwriting Authority. The most consequential cell. Admission to legal-tier underwriting per Lock Box A. Class B Civigens may not select this cell without Class A uplift — an EAAP-POL constraint that prevents underwriting commitments from being made by parties without the attestation depth to honor them. Institutions and investors typically operate in this cell when their core business is underwriting; Civigens uplift to it only when their station genuinely requires it.

§5.7 Failure Modes (Both Frames)

Operational frame failures:

1. **SLA breach.** The underwriting commitment is not honored. Diagnostic: SLA performance audit against contract terms.
2. **Underwriting gap.** A round-trip executes without explicit Infomediator underwriting. Diagnostic: round-trip record check for underwriting attestation.
3. **DID compromise.** A single layer of the Defense-in-Depth tiering fails, and subsequent layers fail to compensate. Diagnostic: DID-tier-by-tier audit.
4. **Hand/Head allocation drift.** The allocation between Hand-tier and Head-tier authorization has shifted away from the Def-Rel rating. Diagnostic: CyberRAVE Def-Rel comparison.

Governance frame failures:

1. **CBGMODD seat vacancy.** A seat is unfilled, leaving the council without one of its seven perspectives. Diagnostic: seat-roll attestation.
2. **CyberRAVE Def-Rel deflation.** The rating substrate's Def-Rel dimension has drifted toward inflated ratings (everyone gets high marks). Diagnostic: rating-distribution audit.
3. **SROC non-redemption.** A Smart-Resonant Offset Contract is invoked but the offset cannot be redeemed. Diagnostic: SROC redemption rate audit.
4. **Alternative EarnedPath instrument designation defect.** An admitted alternative instrument is found to have failed one of the five designation gates retrospectively. Diagnostic: gate-by-gate re-audit.

Operational and governance failures are diagnosed independently but reported together; the doubled frame requires doubled diagnostics.

§5.8 Relationship to EAAP-POL

EAAP-POL constraints on Security-Clearance are heavy because the volume's commitments are restrictive and consequential. Key constraints:

- Class B Civigens may not select Security-Clearance × LAW without uplift to Class A.
- Selection of any Security-Clearance cell requires explicit Infomediator underwriting per Lock Box A.
- Selection of multiple Security-Clearance cells requires CBGMODD seat-coverage attestation (the council seats relevant to the cells must be filled).
- Federation across VERTECA mirrors requires identical Security-Clearance selection profiles or explicit profile-bridging attestation.

EAAP-POL Pass D will specify the constraint set in full; this chapter records the architectural commitments.

Chapter 6 — Data-Integrity

§6.1 Canonical Formulation and Five-Layer Frame

> **Data-Integrity** = SEPLTA == H2C2H / C2H2C === ERES App-Parent 111000111 / 000111000 IPIDITIS / IDIPITIS

Five layers of involutive symmetry (Memo v1.2 §1.3 and Lock Box B):

Layer	Frame	Symmetry
1	SEPLTA admissibility	Six-domain coverage at intake matches at return
2	H2C2H / C2H2C round-trip direction	Bit-complement preservation across direction class
3	ERES App-Parent architecture	Child-app behavior matches parent specification
4	111000111 / 000111000 binary	Palindromic-complement involution preserved
5	IPIDITIS / IDIPITIS character	(2,4) transposition involution preserved

Lock Box B (Memo v1.2 §1.3) constrains the layer count to exactly five; no sixth layer or sub-layer admissible. This chapter unfolds each layer operationally and provides at least one worked counterexample showing how a real payload can break that specific layer while passing the others.

§6.2 Layer 1 — SEPLTA Admissibility

Operational. The qualify stage's six-domain admissibility frame: **S**ocial · **E**cological · **P**olitical · **L**egal · **T**echnical · **A**dministrative. A query is admissible if it fits one or more domains. A query is integrity-symmetric at Layer 1 if its domain coverage at intake (qualify-stage tagging) matches its domain coverage at return (Sim OUTPUT structure). Asymmetry at Layer 1 is the diagnostic: a query tagged as Social/Legal at intake but returned as Technical/Administrative at output has lost its admissibility frame somewhere in the round-trip.

Worked counterexample (Layer 1 break, other layers intact). A Civigen submits a query tagged Social/Ecological at qualify-stage. The DATABANK retrieval returns content that is technically correct, passes round-trip-direction symmetry, conforms to App-Parent, encodes cleanly under bit-complement, and admits (2,4) transposition without semantic loss. But the returned content is exclusively Technical/Administrative — the Social/Ecological framing is lost. Layer 1 fails; Layers 2-5 pass. The Civigen's question has been answered as if it were a technical question, and the loss is not detectable at Layers 2-5. Layer 1 is therefore non-redundant; without it, content can pass the lower-layer integrity checks while drifting away from the qualify-stage admissibility frame.

§6.3 Layer 2 — H2C2H / C2H2C Round-Trip Direction

Operational. The round-trip has two canonical directions:

- **H2C2H** — Human-to-Computer-to-Human. The Civigen submits a query (H→C); the Infomediary processes (C); the Sim OUTPUT returns to the Civigen (C→H). The dominant direction class for Civigen-initiated round-trips.
- **C2H2C** — Computer-to-Human-to-Computer. A computational system submits a query needing human-tier judgment (C→H); the human provides judgment (H); the result returns to computational use (H→C). The dominant direction class for system-initiated round-trips requiring human attestation.

Layer 2 symmetry is preserved when both direction classes produce equivalent DETAIL under bit-complement. That is: the same query, run as H2C2H and run as C2H2C, must produce DETAIL that is bit-complement-equivalent. This is a strong commitment; it ensures the Infomediary does not produce different answers depending on which side of the round-trip initiates.

Worked counterexample (Layer 2 break). A query about EarnedPath action availability runs cleanly as H2C2H and produces DETAIL with action-set X. The same query, initiated by an automated CBGMODD seat-rotation system (C2H2C), produces DETAIL with action-set Y. Bit-complement of X does not equal Y. Layer 1 (SEPLTA) passes both directions; Layers 3-5 pass internally to each direction. But the inter-direction symmetry breaks. The diagnostic: paired H2C2H/C2H2C audit on canonical query corpus.

§6.4 Layer 3 — ERES App-Parent Architecture

Operational. The ERES corpus uses an App-Parent pattern: a parent app specifies structural commitments, and child apps inherit them. The Infomediary itself is a parent; child apps include AD_ON-AI (the composer), the various VERTECA mirrors, the per-deployment-scale instances (THOW-Infomediary, HFVN-Infomediary, FDRV-Infomediary). Child-app behavior must match parent-app specification across the round-trip.

Layer 3 symmetry is preserved when child-app behavior conforms to parent specification. Asymmetry: a child app's behavior diverges from its parent's specification, producing DETAIL that the parent would not have produced.

Worked counterexample (Layer 3 break). AD_ON-AI, as the child composer, receives DETAIL from a parent Infomediary round-trip and composes Sim OUTPUT. The DETAIL is Layer-1-clean (SEPLTA-symmetric), Layer-2-clean (direction-symmetric), and Layer-4/5-clean (binary and character symmetric). But AD_ON-AI's composition introduces phrasing or framing that the parent Infomediary would not have endorsed — perhaps the composer over-fits to a stylistic register, or under-emphasizes a BSG tier the parent specified. The child diverges from the parent. Layer 3 fails.

The App-Parent layer is what catches composer drift. Without it, child apps can stylistically corrupt content that is technically integrity-symmetric.

§6.5 Layer 4 — 111000111 / 000111000 Binary Palindromic-Complement

Operational. The binary representation layer. The two 9-bit sequences `111000111` and `000111000` are individually palindromic (read the same forward and backward) and form a bit-complement pair (each bit flipped). Together they constitute a bit-complement involution: applying the bit-complement to one yields the other; applying it twice returns the original.

Layer 4 symmetry is preserved when the binary encoding of the round-trip's payload admits the bit-complement involution without payload corruption. That is: if you take the payload, encode it in binary, apply bit-complement, decode it, and the result is semantically equivalent to the original — Layer 4 holds. If the bit-complement produces a payload that is semantically distinct or invalid, Layer 4 fails.

Worked counterexample (Layer 4 break). A payload encoded as ASCII produces binary that, when bit-complemented, lands in the high-bit range of UTF-8 encoding and decodes to a different character set entirely. The payload "HELP" (0x48 0x45 0x4C 0x50) bit-complements to (0xB7 0xBA 0xB3 0xAF), which under naive UTF-8 decode produces invalid sequences. Layers 1-3 pass; Layer 5 (character (2,4) transposition) is moot because Layer 4 fails first. The diagnostic: bit-complement round-trip on payload binary.

This counterexample illustrates why Layer 4 is non-redundant: encodings that pass admissibility, direction, and parent-spec checks can still fail at the binary layer if the payload's encoding is not bit-complement-stable.

§6.6 Layer 5 — IPIDITIS / IDIPITIS Character (2,4) Transposition

Operational. The character representation layer. The canonical 8-character sequence IDIPITIS admits a (2,4) transposition: swap the characters at positions 2 and 4. Position 2 of IDIPITIS is D; position 4 is P. Swapping yields IPIDITIS. Applying the transposition twice returns IDIPITIS — the operation is involutive.

Layer 5 symmetry is preserved when the character-level encoding of the round-trip admits the (2,4) transposition without semantic corruption. The payload is character-symmetric at Layer 5 if the (2,4)-transposed version is semantically equivalent.

Worked counterexample (Layer 5 break). A payload contains a string in which the characters at positions analogous to (2,4) carry distinct semantic weight — a financial amount, a coordinate, a cryptographic hash — such that transposition produces a meaningfully different value. Layers 1-4 may pass; Layer 5 fails because the transposition is not semantically invariant.

The IPIDITIS/IDIPITIS pair is the architecture's canonical Layer-5 reference because the transposition between them is well-defined and tractable. Other (2,4)-symmetric payloads inherit the Layer-5 commitment by isomorphism.

§6.7 Non-Extensibility (Lock Box B) Re-Affirmed

Per Lock Box B (Memo v1.2 §1.3): the five layers are exactly five; no sixth layer or sub-layer is admissible. The architecture's claim is that involutive symmetry across these five layers — admissibility, direction, parent-spec, binary, character — is sufficient for Data-Integrity. A sixth layer would introduce asymmetry (5+1 is not involutive at the layer-count level); a sub-layer would fragment one of the existing five and dilute its checking power.

This is not stylistic minimalism. The five-layer structure is mathematically required by the chiasmic-palindromic involution at the architectural level. Pass C may explore why exactly five layers are required (the involutive math); Pass B records the lock as canonical.

§6.8 The Four Data-Integrity Cells

Data-Integrity × HELP — Five-Layer Witness. Five-layer integrity verification applied to help/sustenance interactions. Audit trail of help given and received, witnessed across all five integrity layers. Selected by Civigens who want sustenance interactions to leave clean audit records (medical care, social services, mutual aid).

Data-Integrity × USE — Audit-Trail of Action. Five-layer audit applied to actions taken. The cell where operational action histories are integrity-attested. Selected by parties whose actions need to be defensible under retrospective review (institutions with regulatory exposure, Civigens building EarnedPath records).

Data-Integrity × ENERGY — RATE Composite. BERA RATE composite computed under five-layer integrity. The cell where measurement integrity is most consequential — the BERA reading that is integrity-attested can be relied on; one that is not, cannot. Selected by parties whose decisions depend on BERA accuracy.

Data-Integrity × LAW — Public-Record Attestation. Five-layer-witnessed public-record attestation. The cell where the architecture's wealth-sharing commitment (extracting private epistemology into public record) operates with full integrity guarantees. The most attestation-heavy cell in the matrix; selected by parties whose actions or claims need to enter the public record under audit-grade conditions.

§6.9 Failure Modes (Per Layer)

Each layer fails in a characteristic way:

1. **Layer 1 failure:** Domain-coverage drift between intake and return. *Remedy:* re-qualify and re-run.
2. **Layer 2 failure:** Direction-class divergence. *Remedy:* paired-direction audit and reconciliation.
3. **Layer 3 failure:** Child-parent divergence. *Remedy:* child-app re-attestation against parent spec.
4. **Layer 4 failure:** Binary involution corruption. *Remedy:* re-encode with involution-stable encoding.
5. **Layer 5 failure:** Character transposition semantic break. *Remedy:* payload restructuring to remove transposition-vulnerable positions.

A round-trip that fails any layer returns VEILED at the failed layer with explicit diagnostic and remedy guidance. The Civigen may re-submit after remedy is applied; the round-trip is not silently re-tried.

§6.10 Relationship to EAAP-POL

EAAP-POL constraints on Data-Integrity center on attestation: every Data-Integrity round-trip must produce an attestation record that includes per-layer integrity status (pass/fail/VEILED). Class B Civigens may select Data-Integrity cells freely except Data-Integrity × LAW, which requires Class A uplift for the public-record commitment to be honor-bound.

Part III — The Matrix Walk

§7.1 Reading the Matrix

The 3×4 selection matrix from Memo v1.2 §2.1:

	HELP	**USE**	**ENERGY**	**LAW**
One-Good	sustenance care	adaptive Q&A	resource sufficiency	covenant of provision
Security-Clearance	DID for vulnerable	Hand/Head purchase tier	grid access tier	underwriting authority
Data-Integrity	five-layer witness	audit-trail of action	RATE composite	public-record attestation

Twelve cells. A User-GROUP's selection profile is one or more cell selections. This Part walks each cell once, with operational specification followed by three entry surfaces: **Civigen** (individual or family selecting in their personal capacity), **Institution** (organization selecting in operational capacity), **Investor** (capital allocator selecting in fiduciary capacity). The three surfaces are not exhaustive but are the canonical entries the architecture is designed to serve at v1.x; Pass C will extend to additional entry surfaces (researcher, regulator, adversary).

The walk is mechanical by design. Each cell receives the same structural treatment so that the matrix can be read as a reference rather than a narrative.

§7.2 One-Good × HELP — Sustenance Care

Operational specification. RT lookup of available sustenance resources within reach, anchored to UBIMIA Common Core Sustenance specification. HowWay grounds the answer in canonical material; MyWay particularizes to the Civigen's location, capacity, and immediate need; $IT delivers the concrete access path.

Civigen entry. A single individual or family selects this cell to ensure baseline-need response. Default cell for THOW-tier deployment. Typical query: *"Where can I access [food/medical care/shelter/basic services] within the next [latency window]?"* Sim OUTPUT delivers the access path with HowWay reference and MyWay particularization.

Institution entry. A community center, DOFA family-services office, or mutual-aid organization selects this cell to operationalize their sustenance commitments. Typical query: *"What sustenance demand is in our service area, and what is our current capacity to meet it?"* Sim OUTPUT delivers the demand-capacity mapping.

Investor entry. A philanthropic foundation or impact fund selects this cell to direct capital toward measurable basic-need fulfillment. Typical query: *"What sustenance gaps exist in [region/population], and what capital deployment closes them at what BERA RATE?"* Sim OUTPUT delivers the gap analysis with capital-allocation guidance.

§7.3 One-Good × USE — Adaptive Q&A

Operational specification. RT operationalization of the HowWay/MyWay/$IT triadic Q/A pattern itself. The cell where ordinary day-to-day questions are answered.

Civigen entry. The most-selected cell for single-cell Civigens new to the architecture. Typical query: *"How do I [accomplish task] given [my situation]?"* Sim OUTPUT delivers the personalized method.

Institution entry. Customer-facing or member-facing Q/A platforms; educational institutions; advisory services. Typical query: *"What is the canonical answer to [class of question] for our [population]?"* Sim OUTPUT delivers the institutional Q/A baseline.

Investor entry. Diligence Q/A across portfolio; real-time advisory. Typical query: *"What is the operational status of [portfolio company] on [dimension] given [current conditions]?"* Sim OUTPUT delivers the diligence brief.

§7.4 One-Good × ENERGY — Resource Sufficiency

Operational specification. RT accounting of energy and resource availability against commitment surface.

Civigen entry. Household energy and resource budget tracking. Typical query: *"Are my household energy and resource flows sufficient for the next [period]?"* Sim OUTPUT delivers the sufficiency assessment.

Institution entry. Operational energy/resource management at HFVN or institutional scale. Typical query: *"Are our institutional energy and resource flows on track against commitments?"* Sim OUTPUT delivers the institutional sufficiency assessment.

Investor entry. Resource-flow analysis for portfolio companies and macro-tier investments. Typical query: *"Which portfolio companies are resource-sufficient and which are at risk?"* Sim OUTPUT delivers the portfolio-tier assessment.

§7.5 One-Good × LAW — Covenant of Provision

Operational specification. RT verification of legal commitments to provide. The cell that grounds the architecture's legitimacy in the legal register.

Civigen entry. Civigen rights verification — confirming access to common-core sustenance under DOFA registration. Typical query: *"What sustenance rights does my Civigen registration *entitle me to, and how do I exercise them?"* Sim OUTPUT delivers the rights specification with exercise paths.

Institution entry. Compliance with provision-covenants — legal/charter obligations to provide sustenance to defined populations. Typical query: *"What provision covenants govern our institution, and what is our compliance status?"* Sim OUTPUT delivers the compliance report.

Investor entry. Covenant-compliance audit and regulatory exposure assessment. Typical query: *"Which portfolio companies have provision-covenant exposure, and what is the regulatory risk posture?"* Sim OUTPUT delivers the exposure analysis.

§7.6 Security-Clearance × HELP — DID for Vulnerable

Operational specification. Defense-in-Depth security applied to vulnerable populations: children, elderly, disabled, refugees, those with reduced self-defense capacity. Hand-vs-Head allocation skews to Hand-tier (embodied protection) where vulnerable populations cannot operate Head-tier authorization independently.

Civigen entry. Civigens caring for vulnerable family members select this cell to obtain protection-tier guidance. Typical query: *"What protective measures apply to [vulnerable family member] in [situation], and how do I implement them?"* Sim OUTPUT delivers the DID guidance with Hand-tier emphasis.

Institution entry. Healthcare, child-welfare, eldercare institutions deploy DID at this cell to formalize their protective commitments. Typical query: *"What DID specifications apply to our care population, and what is our current compliance posture?"* Sim OUTPUT delivers the institutional DID specification.

Investor entry. Risk-adjusted investment in protective infrastructure (eldercare facilities, child-welfare technology, refugee-services platforms). Typical query: *"What protective infrastructure investments meet DID-standard for [target population], and what is the BERA-adjusted return profile?"* Sim OUTPUT delivers the investment thesis.

§7.7 Security-Clearance × USE — Hand/Head Purchase Tier

Operational specification. Authorization tiering for transactional use. Hand-tier for routine transactions under threshold; Head-tier for cryptographic-scale (above threshold or cross-jurisdictional).

Civigen entry. Personal financial authorization. Hand-tier for daily transactions; Head-tier for significant or cross-border. Typical query: *"What authorization tier applies to [proposed transaction], and what are the requirements?"* Sim OUTPUT delivers the tier specification with Hand/Head allocation.

Institution entry. Procurement authorization, audit trails, signing authority allocation. Typical query: *"What Hand/Head allocation governs our procurement at [scale], and what attestation is* *required?"* Sim OUTPUT delivers the allocation specification.

Investor entry. Trade authorization tiers, custody arrangements, cryptographic asset management. Typical query: *"What Head-tier authorization governs our cryptographic custody, and what is the Hand-tier fallback?"* Sim OUTPUT delivers the authorization architecture.

§7.8 Security-Clearance × ENERGY — Grid Access Tier

Operational specification. Authorization tiering for energy-grid access across the deployment-scale spectrum. THOW-tier (single-household micro-grid clearance), HFVN-tier (village-network grid clearance), S3C-tier (city-grid clearance), FDRV-tier (closed-grid clearance).

Civigen entry. Grid-tier access at THOW scale. Typical query: *"What grid-access tier applies to my household, and how do I provision authorization?"* Sim OUTPUT delivers the access specification.

Institution entry. Grid integration at HFVN-scale or institutional scale. Typical query: *"How does our institution integrate with the [grid tier], and what authorization architecture is required?"* Sim OUTPUT delivers the integration plan.

Investor entry. Grid-infrastructure investment authorization. Typical query: *"What grid-infrastructure investments at [tier] meet authorization standards, and what is the regulatory profile?"* Sim OUTPUT delivers the investment authorization brief.

§7.9 Security-Clearance × LAW — Underwriting Authority

Operational specification. The most consequential cell. Admission to legal-tier underwriting per Lock Box A. Class B Civigens may not select this cell without Class A uplift.

Civigen entry. Generally not selected. Civigens uplift to it only when their station genuinely requires legal-tier underwriting commitments (a Civigen who has become an Infomediator wielding Infomediaries for others). Typical query (for uplifted Civigen): *"What underwriting commitments are appropriate to my station, and what attestation is required?"* Sim OUTPUT delivers the underwriting specification.

Institution entry. Insurance and underwriting institutions, legal practices, regulatory bodies. The cell where institutional core business engages directly with Lock Box A. Typical query: *"What underwriting capacity does our institution carry, and how is risk allocated under [arrangement]?"* Sim OUTPUT delivers the underwriting architecture.

Investor entry. Underwriting funds, reinsurance, capital pools backing Infomediator commitments. Typical query: *"What is our underwriting exposure across [Infomediators/portfolio], and what is the BERA-attested risk-adjusted return?"* Sim OUTPUT delivers the underwriting portfolio analysis.

§7.10 Data-Integrity × HELP — Five-Layer Witness

Operational specification. Five-layer integrity verification applied to help/sustenance interactions. Audit trail across SEPLTA, direction-class, App-Parent, binary, and character layers.

Civigen entry. Witnessed care interactions. Typical query: *"Maintain a five-layer-witnessed record of [care interaction] for [purpose: medical history, social services, mutual-aid attestation]."* Sim OUTPUT delivers the witness record.

Institution entry. Audit trail of services delivered with integrity guarantees. Typical query: *"Produce five-layer-witnessed audit of services delivered to [population] over [period]."* Sim OUTPUT delivers the witnessed audit.

Investor entry. Verification of impact claims under integrity guarantees. Typical query: *"Verify [portfolio company]'s impact claims under five-layer integrity for [reporting period]."* Sim OUTPUT delivers the verified impact report.

§7.11 Data-Integrity × USE — Audit-Trail of Action

Operational specification. Five-layer audit applied to actions taken.

Civigen entry. Personal action history with integrity preservation. Typical query: *"Maintain a five-layer-attested record of my actions on [matter] for [purpose]."* Sim OUTPUT delivers the attested action history.

Institution entry. Operational audit trails at institutional scale. Typical query: *"Produce five-layer-attested operational audit for [period/scope]."* Sim OUTPUT delivers the operational audit.

Investor entry. Activity audit for portfolio companies under integrity standards. Typical query: *"Audit [portfolio company]'s activity under five-layer integrity for [matter]."* Sim OUTPUT delivers the activity audit.

§7.12 Data-Integrity × ENERGY — RATE Composite

Operational specification. BERA RATE composite computed under five-layer integrity guarantees.

Civigen entry. Energy/resource impact rating with integrity attestation. Typical query: *"Compute my BERA RATE on [activity/commitment] under five-layer integrity."* Sim OUTPUT delivers the integrity-attested RATE.

Institution entry. Institutional BERA scoring with integrity attestation. Typical query: *"Compute our institutional BERA RATE composite under five-layer integrity for [reporting period]."* Sim OUTPUT delivers the institutional RATE.

Investor entry. Portfolio BERA aggregation with integrity attestation. Typical query: *"Aggregate portfolio BERA RATE under five-layer integrity, with per-company attribution."* Sim OUTPUT delivers the portfolio RATE composite.

§7.13 Data-Integrity × LAW — Public-Record Attestation

Operational specification. Five-layer-witnessed public-record attestation. The architecture's wealth-sharing commitment in operational form.

Civigen entry. Legal-tier life events with attestation (births, deaths, marriages, property transfers, contractual commitments). Typical query: *"Attest [life event] to public record under five-layer integrity."* Sim OUTPUT delivers the attested record. Class B Civigens require Class A uplift.

Institution entry. Regulatory filings, public records with attestation. Typical query: *"File [regulatory document] to public record under five-layer integrity."* Sim OUTPUT delivers the attested filing.

Investor entry. Audit-grade investor disclosures. Typical query: *"Produce audit-grade [disclosure] under five-layer integrity for public record."* Sim OUTPUT delivers the attested disclosure.

§7.14 Cross-Cell Composition

A User-GROUP's selection profile is rarely a single cell. Composition across cells is the typical case, and Pass B specifies how cross-cell composition operates.

Same-Volume composition (within a row). Selection of multiple cells within a single Trilogy volume — e.g., `{One-Good × HELP, One-Good × USE, One-Good × ENERGY, One-Good × LAW}` — produces a row-aligned profile. The volume's filter pass runs across all selected cells in the row simultaneously; DETAIL is composited cell-by-cell with shared volume semantics. Same-volume composition is the most operationally tractable case.

Same-Pillar composition (within a column). Selection of multiple cells within a single Pillar — e.g., `{One-Good × LAW, Security-Clearance × LAW, Data-Integrity × LAW}` — produces a column-aligned profile. Multiple volumes run filter passes on the same Pillar. Composition is more expensive than same-volume because the volumes' semantics are heterogeneous; DETAIL is composited per-volume and reconciled at the Pillar level.

Diagonal composition (across both axes). Selection of cells across both volume and Pillar axes — e.g., `{One-Good × HELP, Security-Clearance × USE, Data-Integrity × LAW}` — produces a diagonal profile. The most expensive composition class because multiple volumes run on multiple Pillars; reconciliation is at the matrix level.

Full-matrix composition. Selection of all twelve cells. Approaches FDRV-tier deployment as deployment-scale spectrum (per Pass A FINAL §0.6.1). Full architectural exposure; full architectural cost; full architectural benefit.

The composition cost is borne by AD_ON-AI. The Infomediary itself runs the composition mechanically; the downstream composer (AD_ON-AI) bears the cost of producing Sim OUTPUT that integrates the composited DETAIL into a single coherent return. Composition cost scales with profile complexity. Pass C will specify the cost-bounding diagnostic; Pass B records the architectural commitment.

Profile-tier alignment. Per Memo v1.2 §2.2: selection profiles align to deployment-scale tiers. A THOW-tier profile that is full-matrix is operationally suspicious (over-commitment); an FDRV-tier profile that is single-cell is operationally suspicious (under-commitment). Misalignment is an auditable diagnostic, not an architectural failure; Pass B Part III §7.14 records the diagnostic, and Pass C §11 will specify the resolution path.

Part IV — DATABANK Architecture under DOFA Governance

§8.1 The DATABANK Specification

The DATABANK formula from Pass A FINAL §0.5:

> **ERES DATABANK = Document Library × UBIMIA-Manual + Other-Ref**

Three components, each load-bearing.

Document Library. The 600+ documents constituting the ERES corpus as of May 2026. ResearchGate publications (430+), GitHub repositories under ERES-Institute-for-New-Age-Cybernetics, the Hague filing materials (March 2026), the Berggruen Prize Essay, the closed working drafts, and the slot 431-434 cluster. The Library is the body of canonical material the Infomediary draws from.

UBIMIA-Manual. The v1.1 Manual (April 25, 2026, 188 pages) plus its companion architecture. The Manual specifies the Constitutional Pillars, the BERA architecture, the CBGMODD seven-seat council, SCALULAR, and the Department of Family Amity. The Manual is the spine — the constitutional framework that shapes how the Library's body is read.

Other-Ref. The named-but-undefined extension point. Reserved for future corpora — third-party reference materials, peer-reviewed external work, ratified amendments, post-2026 expansion. Other-Ref is honest placeholder for what does not yet exist.

The cross-product `Document Library × UBIMIA-Manual` is the Library *as filtered through* the Manual's constitutional commitments. The Library alone is corpus; the Manual alone is constitutional spine; their cross-product is the operationally-useful DATABANK content.

§8.2 DOFA Seven-Seat Council as Governance Interface

Per Pass A FINAL §0.9 and Memo v1.2 §2.3, DOFA governs the Infomediary. The DATABANK, as the Infomediary's queryable corpus, is therefore governed by DOFA. The seven-seat CBGMODD council is DOFA's operational expression and constitutes the DATABANK's governance interface.

Each seat carries a distinct governance perspective on the DATABANK:

- **Citizen seat** — represents the broadest population class. Gates DATABANK content for accessibility, comprehensibility, and Civigen-tier relevance.
- **Business seat** — represents commercial activity. Gates content for commercial viability, market alignment, and operational tractability.
- **Government seat** — represents public authority. Gates content for legal compliance, regulatory alignment, and public-interest framing.
- **Military seat** — represents defense and resilience. Gates content for security implications, threat-model coverage, and resilience commitments.
- **Ombudsman seat** — represents audit and redress. Gates content for transparency, audit-trail completeness, and dispute-resolution paths.
- **Dignitary seat** — represents spiritual and cultural register. Gates content for moral vectoring, cultural respect, and spiritual coherence.
- **Diplomat seat** — represents cross-federation interface. Gates content for VERTECA-mirror compatibility, jurisdictional translatability, and federation-level reciprocity.

Seven seats, seven perspectives. DATABANK content is admitted when all seven seats endorse (or, under specified policy, when a supermajority endorses with named-seat dissent recorded). Content that fails any seat's gate is held — not deleted, but quarantined for review and potential remediation.

The seven-seat governance is what makes the DATABANK *trustworthy*. A corpus governed by a single perspective is a vulnerability; a corpus governed by seven non-redundant perspectives is robust to single-perspective drift.

§8.3 UBIMIA-Manual as Constitutional Spine

The UBIMIA Manual v1.1 is the DATABANK's constitutional spine in the strongest sense: it specifies what content the DATABANK *must* contain, what content it *may* contain, and what content it *may not* contain.

Must contain. Specifications of the Constitutional Pillars (HELP, USE, ENERGY, LAW) and their operational commitments; BERA architecture and the seven RATE dimensions; CBGMODD seven-seat council structure and seat default profiles; SCALULAR specifications across the four pillars; DOFA structure and operational commitments. These are non-optional content.

May contain. The 600+ Document Library entries, the slot 431-434 cluster materials, the Berggruen Prize Essay, the Hague filing materials, the closed working drafts, and any future

ResearchGate publications. These are admitted contingently — admission is governed by the seven-seat council per §8.2.

May not contain. Content that violates the architecture's moral vectoring (BEST/SOUND/GOOD calibration); content that fails Lock Boxes A, B, or C; content that introduces auto-routing semantics in violation of Lock Box C; content that adds layers to Data-Integrity in violation of Lock Box B; content that asserts underwriting direction other than per Lock Box A. These are non-negotiable exclusions.

The Manual's spine function is what makes the cross-product `Document Library × UBIMIA-Manual` operationally meaningful. Without the spine, the Library is a corpus without commitments; with the spine, the Library is a corpus that knows what it is.

§8.4 Other-Ref as Extension Point

Other-Ref is the DATABANK's commitment to its own future. The architecture is designed to last a thousand years; corpora that bind themselves to specific external dependencies do not last that long. Other-Ref is the named-but-undefined slot through which future corpora can be admitted without re-architecting the DATABANK.

Pass B specifies the **admission process** for Other-Ref content:

1. **Proposal stage.** A party (Civigen, Institution, Investor, or external researcher) proposes admission of an external corpus to Other-Ref. The proposal includes: the corpus identity, the relevance argument, and the licensing-compatibility assessment.
2. **Seven-seat review.** The CBGMODD council reviews the proposal under the same process as Document-Library admission (§8.2).
3. **Pilot integration.** Approved proposals are integrated under a pilot designation that bounds their DATABANK influence to non-canonical query support.
4. **Promotion or rejection.** After pilot period, the corpus is either promoted to canonical Other-Ref status or rejected with named-seat dissent recorded.
5. **Periodic re-review.** All canonical Other-Ref content is re-reviewed at specified intervals (Pass D will specify the interval) for continued relevance and licensing compliance.

Other-Ref is therefore not a backdoor; it is a deliberately-designed extension mechanism with the same governance discipline as the rest of the DATABANK.

§8.5 The App-Parent Pattern as Architectural Lock

Per Pass A FINAL §0.5 and Pass B §6.4, the DATABANK is governed by the App-Parent pattern. The DATABANK itself is the parent corpus; the per-deployment-scale instances (THOW-DATABANK, HFVN-DATABANK, FDRV-DATABANK) are children that inherit the parent's structural commitments.

Child-instance commitments inherited from parent DATABANK:

- The three-component structure (Library × Manual + Other-Ref).
- The seven-seat governance interface.
- The cell-bounded access guarantee (per Lock Box C).
- The five-layer Data-Integrity attestation availability (per Lock Box B).
- The underwriting-direction commitments (per Lock Box A) for Security-Clearance interactions.

Child instances may adapt their content to their deployment scale (a THOW-DATABANK is a smaller subset of the parent than an HFVN-DATABANK), but they may not violate the parent's structural commitments. A child that does so is, per Pass B §6.4 (Layer 3), Data-Integrity-defective.

The App-Parent pattern is what allows the DATABANK to deploy at THOW, HFVN, and FDRV scales without architectural fragmentation. The parent specifies once; children inherit once; the architecture stays coherent across the deployment-scale spectrum.

§8.6 GSSG Substrate as Material Foundation

Per Pass A FINAL §0.6.2 and Memo v1.2 §1.4, GSSG (Graphene-Sugar Self-healing Grid) is the material substrate underpinning all deployment scales. The DATABANK, as the Infomediary's queryable corpus, has a material expression: it is stored, accessed, and propagated on physical infrastructure. That infrastructure rides on GSSG.

The material/operational isomorphism (Memo v1.2 §1.4) implies a specific commitment for the DATABANK: **DATABANK persistence at any deployment scale is materially self-healing.** Storage failures, transmission corruption, federation-mirror drift — all of these have material expressions, and all of them are recoverable through the same involutive feedback that the operational architecture uses.

Concretely:

- **Storage layer.** GSSG-grade storage media admit physical self-healing; corrupted bits in the binary layer (Layer 4 of Data-Integrity) recover through molecular re-association.
- **Transmission layer.** GSSG-grade transmission infrastructure admits self-healing of transient corruption; Layer 2 (round-trip direction) is preserved across material disturbance.
- **Federation layer.** VERTECA mirrors built on GSSG inherit the self-healing property across mirror reconciliation; cross-mirror DATABANK divergence is materially recoverable.

The DATABANK's claim to thousand-year persistence is materially grounded in GSSG. Pre-v1.2, this commitment was implicit (the architecture intended to last but did not name the material substrate); post-v1.2, the commitment is explicit and the substrate is named.

§8.7 The DATABANK at Deployment Scale

Combining §§8.1-8.6: the DATABANK manifests at each deployment scale (THOW, HFVN, FDRV) as an App-Parent child instance, materially embodied in GSSG, governed by the seven-seat CBGMODD council, anchored to the UBIMIA Manual constitutional spine, drawing from the Document Library cross-product, with Other-Ref reserved for future expansion.

A THOW-DATABANK is the minimum viable instance — the corpus a single household can carry materially, the governance interface that a single household can operate (which means the seven-seat council operates by DOFA-delegated authority rather than household-internal seven seats), and the access patterns that single-household selection profiles exercise.

An HFVN-DATABANK is the federation-tier instance — the corpus that a coordinated village can maintain, with seven-seat governance becoming operationally meaningful at this scale (as per Pass A FINAL §0.6.1), and access patterns that span multiple households' coordinated profiles.

An FDRV-DATABANK is the maximum-scale instance — the corpus that a generation-ship-class vessel must carry self-contained, with full seven-seat governance, full UBIMIA-Manual constitutional spine, and access patterns approaching full-matrix selection profiles.

The DATABANK is one DATABANK, expressed at three deployment scales, on one material substrate, under one governance interface. This is what makes the architecture thousand-year-coherent.

Pass B Closure

Part II, Part III, and Part IV are draft. Pass B does not advance to FINAL until:

- The Pass B representative queries are validated against authentic Civigen / Institution / Investor sources (not synthesized as in the Pass A FINAL §1.1 case).
- The Layer-by-Layer counterexamples in Chapter 6 are reviewed for technical accuracy by the MIEVM ensemble.
- The CBGMODD seven-seat default profiles (Pass A FINAL §3 benefit #4, Memo v1.2) are specified — Pass B references them but does not define them in full.
- The matrix walk's three entry surfaces are reviewed for completeness; Pass C may add researcher / regulator / adversary surfaces.

Pass C targets (after Pass B FINAL):

- **Part V** — BEE-NEXUS and Children-of-All-Ages (the architecture's intergenerational commitment expressed in NEXUS terms)

- **Part VI** — The PlayNAC TABLE Sub-Domain Bridges (the cross-domain commitments that allow PlayNAC to integrate)
- **Part VII** — Failure Modes and the VEILED State (the architecture's account of its own failures, the resolution paths, the diagnostic commitments)

Pass D targets (after Pass C FINAL):

- **Part VIII** — Governance, Versioning, Covenant
- Back Matter — Glossary, References, Slot Reservations, License, MIEVM Synthesis Note, Index
- v2.0 integration sweep — merging Pass A FINAL v1.2, Pass B FINAL, Pass C FINAL, and Pass D into a single canonical Volume Zero of the ERES Trilogy.

> Sprute, J. A. (2026). *The Infomediary: The Instrument-of-Faith for the 1000-Year Future Map (Pass B Draft)*. ERES-INFOMEDIARY-BOOK-2026-001-v1.3-PASS-B-DRAFT. ERES Institute for New Age Cybernetics.

> *Don't hurt yourself. Don't hurt others. Build for generations to come.*
> — Joseph A. Sprute, ERES Institute for New Age Cybernetics

— End of Pass B Draft —